Words to equations

1 For each of the following word statements:

i Write the word statement as an equation, where p is the unknown number.

ii Solve the equation.

a The sum of an unknown number and 39 is 83.

b 75 is equal to the sum of an unknown number and 42.

c The product of 3 and an unknown number is 18.

d -7 multiplied by an unknown number is 84.

e 17 less than an unknown number is 25.

f The quotient of an unknown number and 8 is -12.

g 2 times the sum of an unknown number and 5 is 14.

h The sum of a number and 2 divided by 3 is 6.

2 A bag of lollies contains 80 lollies. The lollies are shared evenly between 10 children.

Let n represent the number of lollies each child receives.

a Write down the equation that represents the relationship between the total number of lollies and the number of lollies each child receives.

b Hence, solve the equation.

3 Justin divides a deck of cards into 7 even piles. There are 12 cards in each pile.

Let c represent the total number of cards in the deck.

a Write down the equation that represents the relationship between the total number of cards in the deck and the number of cards in each pile.

b Hence, solve the equation.

4 Tricia gives Tom 25 marbles from her collection. Tricia now has only 44 marbles in her collection.

Let m represent the number of marbles she had before giving some away.

a Write down the equation that represents the relationship between the initial number of marbles and the number of marbles left in her collection.

b Hence, solve the equation.

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5 Lisa bought a hat and a dress for a total of $\$80$. Let h represent the cost of the hat and d represent the cost of the dress.

a Write down the equation that represents the relationship between the total cost and the individual costs of the hat and dress.

b Hence, find the cost of the hat if the cost of the dress is $\$56$.

Tables

6 Consider the word statement "y is equal to the product of 7 and x".

a Write an equation in the form $y = \ldots$ that describes the word statement.

b Hence, complete the table.

x	1	2	3	4	5
y					

7 Consider the word statement "v is equal to the product of -8 and u".

a Write an equation in the form $v = \ldots$ that describes the word statement.

b Hence, complete the table.

u	1	2	3	4	5
v					

8 Consider the word statement "q is equal to p less than 12".

a Write an equation in the form $q = \ldots$ that describes the word statement.

b Hence, complete the table.

p	-2	-1	0	1	2
q					

9 Consider the word statement "19 subtracted from r is equal to s".

a Write an equation in the form $s = \ldots$ that describes the word statement.

b Hence, complete the table.

r	-2	-1	0	1	2
s					

10 Consider the word statement "y is equal to the product of 3 and the sum of x and 4".

a Write an equation in the form $y = \ldots$ that describes the word statement.

b Hence, complete the table.

x	1	2	3	5	10
y					

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11 Consider the word statement "v is equal to the product of 5 and the sum of u and 2".

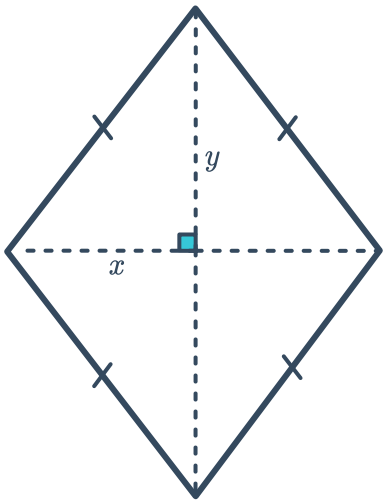
a Write an equation in the form $v = \ldots$ that describes the word statement.

b Hence, complete the table.

u	1	2	3	5	10
v					

Formulas

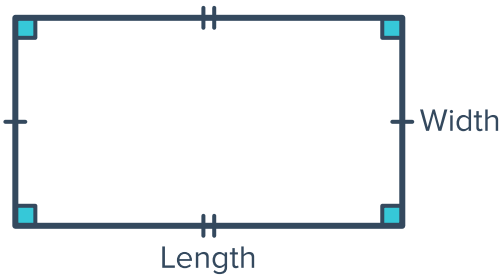
12 The area of a rhombus is given by the formula $A = \frac{1}{2}xy$, where x and y are the lengths of the diagonals:



a Find the area of a rhombus which has short and long diagonal lengths of 4 cm and 6 cm respectively.

b A particular rhombus has a short diagonal length $x = 6\text{ m}$ and area $A = 33\text{ m}^2$. Find the value of y.

13 The area of a rectangle is given by the formula $A = \text{length} \times \text{width}$:

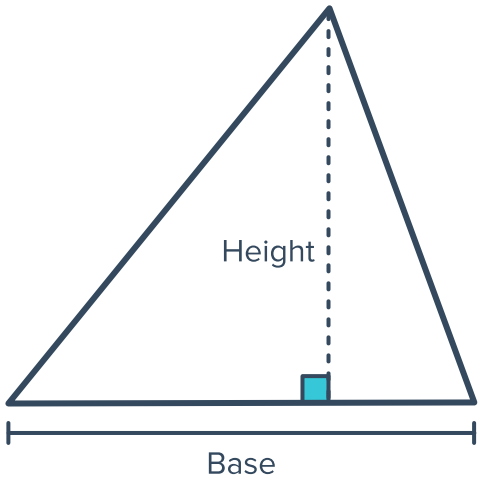


a If the length of a rectangle is 9 m and its width is 3 m, find its area.

b The area of a particular rectangle is 45 m^2. If its length is 9 m, determine the width of the rectangle.

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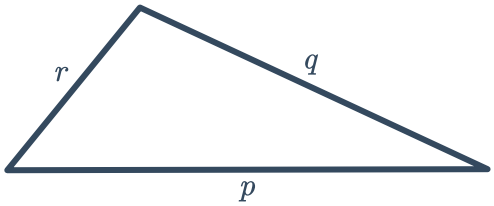
14 The area of a triangle is given by the formula $A = \frac{1}{2} \times$ base $\times$ height:



a If the base of a triangle is 8 mm and its height is 6 mm, find its area.

b A triangle with a height of 4 mm has an area of 16 mm^2. Find the length of the base of the triangle.

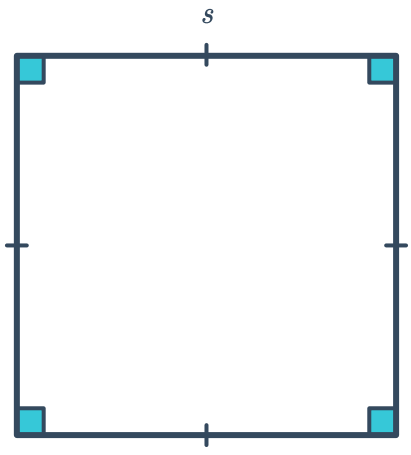
15 The perimeter of a triangle with sides of lengths p, q and r is given by the formula $P = p + q + r$:



a Find P if the length of each of its three sides are $p = 9$ mm, $q = 7$ mm and $r = 5$ mm.

b If $P = 40$ mm and the length of each of the two sides are $p = 10$ mm and $q = 12$ mm, what is the length of the third side r?

16 The perimeter of a square with side lengths of s is given by the formula $P = 4 \times s$:

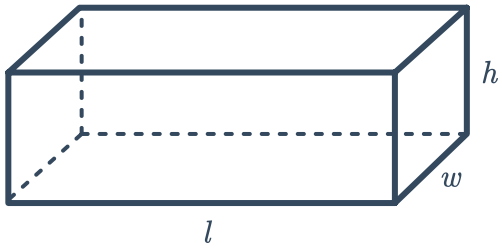


a Find P if the length of each side is 3 m.

b Find s if the perimeter is $P = 40$m

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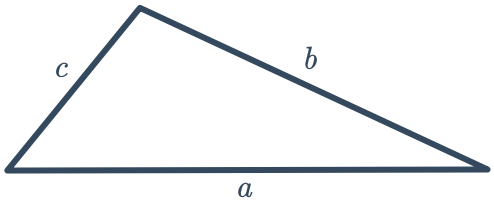
17 The volume of a rectangular prism is given by the formula $V = l \times w \times h$, where l , w and h are the dimensions of the prism:



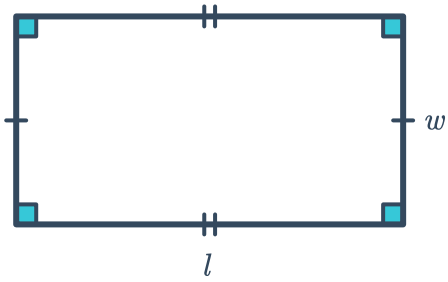
a Given that a rectangular prism has a length of $9\ \text{m}$, a width of $4\ \text{m}$ and a height of $7\ \text{m}$, find its volume.

b The volume of a particular rectangular prism is $140\ \text{cm}^3$. If it has a length of $10\ \text{cm}$ and a width of $7\ \text{cm}$, find the height, h.

18 The perimeter of a triangle can be calculated with the formula $P = a + b + c$, where a, b and c are the three side lengths of the triangle. If a triangle has a perimeter of $P = 37\ \text{cm}$ and side lengths $a = 14\ \text{cm}$ and $b = 13\ \text{cm}$, find the length of the side c.



19 The perimeter of a rectangle is given by the formula $P = 2 \times (l + w)$, where l is the length and w is the width:



a Find the perimeter of the rectangle which has a length of $l = 6\ \text{cm}$ and width of $w = 5\ \text{cm}$.

b If a certain rectangle has length $l = 7\ \text{cm}$ and perimeter $P = 24\ \text{cm}$, find the width w.

20 The speed of a plane can be calculated using the formula $S = \dfrac{D}{T}$, where D is the distance travelled, T is the time taken and S is the speed.

a If a plane travels 3600 kilometres in 6 hours, find its speed.

b If the speed of the plane is $700\,\text{km/hr}$, find the distance travelled after 4 hours.

21 The formula to convert temperature from Celsius to Fahrenheit is $F = 32 + \dfrac{9C}{5}$.

a If $C = 35$, find the value of F.

b If $F = 212$, find the value of C.

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